

Reflection: Probabilistic Models

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The Impact of Fully Bayesian Methods on Decision Making

Concrete Example from Assignment: When evaluating the churn rate for the newly launched, month-to-month contract tier (**Group A_small**, $n = 40$), the frequentist Maximum Likelihood Estimate (MLE) gave a point estimate of **37.50%** (15 churns out of 40). However, the fully Bayesian approach—using a domain-informed **Beta(2, 8)** prior—yielded a Maximum A Posteriori (MAP) estimate of **33.33%** and a 94% Highest Density Interval (HDI) of **[0.210, 0.473]**.

How it Changes the Decision: If I had relied solely on the 37.50% MLE, the business decision would have been to immediately launch an aggressive, high-cost retention campaign targeted at this new segment, as a 37.5% churn rate appears dangerously high compared to the company average.

However, the fully Bayesian answer changes this decision to a "wait and see" approach, opting to hold off on the expensive campaign and collect more data. The 94% HDI explicitly shows that at a 94% credible level, the true churn rate could easily be as low as 21.0% (which is below the overall company average of ~26.5%). By observing this wide interval, it becomes clear that the high MLE is likely just a small-sample artifact rather than a true retention crisis.

The Mechanism: This decision shift is driven by two key Bayesian mechanisms:

1. **Uncertainty Quantification:** The frequentist MLE provides only a point estimate with no measure of confidence for this small sample. The Bayesian 94% HDI **[0.210, 0.473]** explicitly quantifies the severe uncertainty in the data, preventing an overreaction to noise.
2. **Prior Information (Regularization):** The **Beta(2, 8)** prior acts as a regularizer. Because the sample size ($n = 40$) is too small to completely overwhelm the prior beliefs, the prior "pulls" the estimate down from 37.50% to a more conservative 33.33% (a prior pull of $|\text{MAP} - \text{MLE}| = 0.0417$). This tempers the extreme frequentist estimate with established domain knowledge, leading to a more measured, cost-effective business response.